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Academic year 2024-2025 (ODD Sem)**DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT**

Date	25th November 2024	Maximum Marks	10 + 50
Course Code	IM352IA	Duration	120 Min
Sem	V	CIE – I	
OPERATIONS MANAGEMENT			

Note:

1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
Part – A				
1	Operations ensure the efficient use of _____ to meet customer demands.	01	1	1
2	A supply chain includes the flow of materials, information, and _____ across organizations.	01	1	1
3	Effective supply chain management ensures timely delivery and reduces _____.	01	1	1
4	Competitive priorities like cost, quality, and _____ determine how a company competes in the market.	01	1	1
5	The Fourth Industrial Revolution integrates technologies like IoT, _____, and robotics into operations.	01	1	1
6	Processes can be categorized as _____ or supporting based on their direct impact on the final product.	01	1	1
7	Differentiate between order winner and order qualifier.	02	2	2
8	A hospital is considering a new procedure to be offered at Rs.200 per patient. The fixed cost per year would be Rs.100,000, with total variable costs of Rs.100 per patient. What is the break-even quantity for this service?	02	2	2
Part –B				
1	Define operations management. Briefly explain the key terms involved in the definition of operations management with a schematic model of a production system.	10	2	1
2	With suitable example, Distinguish between Manufacturing and service processes.	10	3	1



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Academic year 2024-2025 (ODD Sem)

3	White Valley Ski Resort is planning the ski lift operation for its new ski resort. Management is trying to determine whether one or two lifts will be necessary; each lift can accommodate 250 people per day. Skiing normally occurs in the 14-week period from December to April, during which the lift will operate 7 days per week. The first lift will operate at 90 percent capacity if economic conditions are bad, the probability of which is believed to be about a 0.3. During normal times the first lift will be utilized at 100 percent capacity, and the excess crowd will provide 50 percent utilization of the second lift. The probability of normal times is 0.5. Finally, if times are really good, the probability of which is 0.2, the utilization of the second lift will increase to 90 percent. The equivalent annual cost of installing a new lift, recognizing the time value of money and the lift's economic life, is \$50,000. The annual cost of installing two lifts is only \$90,000 if both are purchased at the same time. If used at all, each lift costs \$200,000 to operate, no matter how low or high its utilization rate. Lift tickets cost \$20 per customer per day. Should the resort purchase one lift or two?	20	4	3															
4	A manager is deciding whether to build a small or a large facility. Much depends on the future demand that the facility must serve, and demand may be small or large. The manager knows with certainty the payoffs that will result under each alternative, shown in the following payoff table. What is the best alternative for each decision rule: Maximin. Maximax. Laplace. Minimax Regret. <table><tr><td></td><td colspan="2">Possible future demand</td></tr><tr><td>Alternative</td><td>Low</td><td>High</td></tr><tr><td>Small facility</td><td>200</td><td>270</td></tr><tr><td>Large facility</td><td>160</td><td>800</td></tr><tr><td>Do nothing</td><td>0</td><td>0</td></tr></table>		Possible future demand		Alternative	Low	High	Small facility	200	270	Large facility	160	800	Do nothing	0	0	10	4	3
	Possible future demand																		
Alternative	Low	High																	
Small facility	200	270																	
Large facility	160	800																	
Do nothing	0	0																	



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Academic year 2024-2025 (ODD Sem)**DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT**

Date	26th November 2024	Maximum Marks	10 + 50
Course Code	IM3531A	Duration	120 Mins
Sem	V Semester	CIE – I	
QUALITY ASSURANCE			

Note:

1. Answer all the Questions.

PART – A

Sl. No	Questions	M	BT	CO
1	Differentiate between defect and defective.	2	2	1
2	What is the primary goal of quality assurance?	1	2	1
3	How does ISO certification benefit organizations?	1	1	2
4	Mention four key core technologies that drive Quality 4.0.	2	2	3
5	List the 7 QC tools	2	1	3
6	What is ATS and ARL?	2	1	2

PART – B

Sl. No	Questions	M	BT	CO
1.	A certain type of storage of battery lasts on an Average 3 years and Standard of 0.5, assume normal distribution, find percentage of battery that lasts a) Less than 2 years, b) Less than 4 years, c) between 2.5 to 4.5 years, d) less than 4 years.	10	3	3
2.	Describe the principles of Rational Subgrouping and apply them to a specific industry, such as manufacturing or service, to show how subgrouping can be used to reduce variation and improve process quality.	10	2	1
3.	Classify the Cost of Good Quality and Cost of Bad Quality in the automotive industry. Provide examples of each cost category and explain how they can be managed effectively to improve product performance and reduce overall expenses.	10	1	2
4.	Choose a specific product (e.g., a smartphone, automobile, or any consumer electronics) and analyze how the Eight Dimensions of Quality influence its acceptance by customers. Break down each dimension—such as performance, features, reliability, and serviceability—and evaluate their relative importance in shaping customer perceptions. How do these dimensions interact with each other to create a competitive advantage for the product in the market? Use relevant examples to support your analysis.	10	4	4
5.	What are the key benefits of implementing the ISO 14000 standard for an organization?	10	2	2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks



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Academic year 2024-2025 (ODD Sem)

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Max Marks	Quiz	03	06	04	—	—	05	05	—	—	—	—
		Test	10	20	10	10	—	10	20	10	10	—	—



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Academic year 2024-2025 (ODD Sem)**DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT**

Date	26th November 2024	Maximum Marks	10 + 50
Course Code	IM254TA	Duration	120 Mins
Sem	V Semester	CIE – I	
FINANCIAL ACCOUNTING AND COSTING			

Note:

1. Answer all the Questions.

PART – A

Sl. No	Questions	M	BT	CO
1	Write a note on double-entry bookkeeping.	02	2	1
2	List the main financial statements prepared in financial accounting	02	1	1
3	Mention the accounting equation	02	1	1
4	What is the purpose of a trial balance?	02	2	1
5	Depict the journal format in accounting.	02	1	1

PART – B

Sl. No	Questions	M	BT	CO
1.	Explain the steps involved in the accounting cycle and discuss their significance in ensuring accurate financial reporting	10	3	1
2.	Discuss how zero-based budgeting differs from traditional budgeting approaches. What are the benefits and limitations of adopting a zero-based budgeting system?	10	2	1
3 a.	Distinguish between Single Entry vs. Double Entry accounting process.	05	2	1
b.	Write a note on International Financial Reporting Standards.	05	2	1
4 a.	Journalise the business transactions during the month of Jan 2024. Jan 1: Owner invested ₹50,000 cash in the business. Jan 3: Purchased office furniture for ₹10,000, paid in cash. Jan 5: Bought goods worth ₹20,000 on credit from M/s XYZ Traders. Jan 10: Paid ₹5,000 to M/s XYZ Traders as partial payment. Jan 15: Sold goods for ₹15,000 on cash. Jan 20: Paid office rent of ₹3,000 in cash.	05	3	2
b.	Discuss the principles of debit and credit in journalizing transactions. How do these principles apply to different types of accounts?	05	3	1



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Academic year 2024-2025 (ODD Sem)

5.	Journalise the business transactions during the month of Feb XXXX Feb 2: Started business with cash ₹1,00,000 and machinery ₹50,000. Feb 5: Opened a bank account by depositing ₹30,000. Feb 10: Purchased goods for ₹40,000 on credit from M/s ABC Traders. Feb 12: Paid ₹5,000 for office rent in cash. Feb 17: Sold goods for ₹25,000 on credit to M/s XYZ Ltd. Feb 20: Paid ₹10,000 to M/s ABC Traders. Feb 22: Purchased office supplies for ₹2,000 in cash. Feb 24: Received ₹20,000 from M/s XYZ Ltd. Feb 27: Paid wages ₹4,000 in cash. Feb 28: Withdrew ₹3,000 for personal use.	10	3	2
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BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5
	Max Marks	Quiz	10	—	—	—	—	06	04	—	—	—
		Test	35	15	—	—	—	—	20	30	—	—



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Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	27 th November 2024	Maximum Marks	10 + 50
Course Code	IM255TBB	Duration	120 Mins
Sem	V	CIE – I	
ENTERPRISE INFORMATION SYSTEMS (Elective)			

Note:

1. Answer all the Questions.

PART – A

Sl. No	Questions	M	BT	CO
1	List two key features of a Customer Relationship Management (CRM) system.	02	L2	CO1
2	Mention two types of data commonly handled by EIS.	02	L2	CO1
3	Name two popular ERP software solutions.	02	L1	CO2
4	What is the primary function of Supply Chain Management (SCM) within EIS?	02	L1	CO1
5	What are the core modules of an ERP system?	02	L2	CO2

PART – B

Sl. No	Questions	M	BT	CO
1.	Analyse the benefits and challenges of implementing an EIS in a manufacturing organization. Support your answer with examples.	10	L2	CO1
2.	How can information technology support a company's business processes and decision making and give it competitive advantage? Give examples to illustrate your answer.	10	L3	CO3
3.	Highlight some of the issues you have faced in using SAP implemented in your institution. How you resolved those issues.	10	L3	CO2
4.	Evaluate the role of EIS in Customer Relationship Management (CRM) with specific examples of how companies have leveraged EIS for CRM.	10	L2	CO3
5.	XYZ Manufacturing Company has been facing inefficiencies in its operations due to a lack of integration between its departments. The procurement, inventory, production, and sales teams operate on standalone systems, leading to delays, data inaccuracies, and increased costs. To address these issues, the company has decided to implement an Enterprise Resource Planning (ERP) system. However, after six months of implementation, the company is still struggling with system adoption and is not seeing the expected improvements. Analyze the challenges XYZ Manufacturing is facing in implementing the ERP system. Suggest strategies to overcome these challenges and ensure successful adoption of the ERP system.	10	L4	CO4

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks



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Academic year 2024-2025 (ODD Sem)

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Max Marks	Quiz	06	04	—	—	—	04	06	—	—	—	—
		Test	10	10	20	10	—	—	20	20	10	—	—



Academic year 2024-2025 (ODD Sem)

**DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT**

Date	27th November 2024	Maximum Marks	50
Course Code	IM355TBD	Duration	90 Min
Sem	VI	CIE – 1	
ADVANCED DECISION MODELLING (Professional Core Elective)			

Note:

1. Answer all the Questions.

Sl. No	Questions	M	BT	CO																																
Part A																																				
1.	What is the primary objective of sequencing problems?	02	1	1																																
2.	How does the concept of ‘setup time’ impact sequencing problems?	02	3	3																																
3.	What is the significance of ‘priority rules’ in sequencing problems?	02	2	2																																
4.	What is the difference between ‘flow shop’ and ‘job shop’ sequencing?	02	2	2																																
5.	How does the concept of ‘setup time’ impact sequencing problems?	02	3	3																																
Part B																																				
1	Explain the following terminologies: i. Idle time on machine ii. Total elapsed time iii. No passing rule iv. Number of machines v. Processing order	10	1	1																																
2	Find the sequence that minimizes the total elapsed time (in hours) required to complete the following tasks on two machines. <table><tr><td>Task</td><td>A</td><td>B</td><td>C</td><td>D</td><td>E</td><td>F</td><td>G</td><td>H</td><td>I</td></tr><tr><td>Machine I</td><td>2</td><td>5</td><td>4</td><td>9</td><td>6</td><td>8</td><td>7</td><td>5</td><td>4</td></tr><tr><td>Machine II</td><td>6</td><td>8</td><td>7</td><td>4</td><td>3</td><td>9</td><td>3</td><td>8</td><td>11</td></tr></table>	Task	A	B	C	D	E	F	G	H	I	Machine I	2	5	4	9	6	8	7	5	4	Machine II	6	8	7	4	3	9	3	8	11	10	2	2		
Task	A	B	C	D	E	F	G	H	I																											
Machine I	2	5	4	9	6	8	7	5	4																											
Machine II	6	8	7	4	3	9	3	8	11																											
3	Determine the optimal sequence of jobs that minimize the total elapsed time, based on the following information. Processing time on machines is given in hours and passing is not allowed. <table><tr><td>Job No</td><td>A</td><td>B</td><td>C</td><td>D</td><td>E</td><td>F</td><td>G</td></tr><tr><td>Machine M₁</td><td>3</td><td>8</td><td>7</td><td>4</td><td>9</td><td>8</td><td>7</td></tr><tr><td>Machine M₂</td><td>4</td><td>3</td><td>2</td><td>5</td><td>1</td><td>4</td><td>3</td></tr><tr><td>Machine M₃</td><td>6</td><td>7</td><td>5</td><td>11</td><td>5</td><td>6</td><td>12</td></tr></table>	Job No	A	B	C	D	E	F	G	Machine M ₁	3	8	7	4	9	8	7	Machine M ₂	4	3	2	5	1	4	3	Machine M ₃	6	7	5	11	5	6	12	10	3	3
Job No	A	B	C	D	E	F	G																													
Machine M ₁	3	8	7	4	9	8	7																													
Machine M ₂	4	3	2	5	1	4	3																													
Machine M ₃	6	7	5	11	5	6	12																													



Academic year 2024-2025 (ODD Sem)

4	There are four jobs each of which has to be processed on machines A, B, C, D, E and F in the order ABCDEF. Processing time in hours is given in the table below. Find the optimal sequencing of jobs, minimum time required to process these jobs and the idle time for each of the machines. <table><tr><td>Job</td><td>A</td><td>B</td><td>C</td><td>D</td><td>E</td><td>F</td></tr><tr><td>1</td><td>15</td><td>8</td><td>6</td><td>14</td><td>6</td><td>36</td></tr><tr><td>2</td><td>17</td><td>7</td><td>9</td><td>10</td><td>15</td><td>22</td></tr><tr><td>3</td><td>21</td><td>7</td><td>12</td><td>9</td><td>11</td><td>19</td></tr><tr><td>4</td><td>18</td><td>6</td><td>11</td><td>12</td><td>14</td><td>17</td></tr></table>	Job	A	B	C	D	E	F	1	15	8	6	14	6	36	2	17	7	9	10	15	22	3	21	7	12	9	11	19	4	18	6	11	12	14	17	10	2	2
Job	A	B	C	D	E	F																																	
1	15	8	6	14	6	36																																	
2	17	7	9	10	15	22																																	
3	21	7	12	9	11	19																																	
4	18	6	11	12	14	17																																	
5	Two major parts P_1 and P_2 for a product require processing through machine centres. The technological sequence of the parts on six machines and manufacturing times on each machine are <table><tr><td rowspan="2">Part P_1</td><td>Machine sequence:</td><td>C</td><td>A</td><td>E</td><td>F</td><td>D</td><td>B</td></tr><tr><td>Time (hrs):</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>1</td></tr><tr><td rowspan="2">Part P_2</td><td>Machine sequence:</td><td>B</td><td>A</td><td>E</td><td>F</td><td>C</td><td>D</td></tr><tr><td>Time (hrs):</td><td>3</td><td>2</td><td>5</td><td>3</td><td>2</td><td>3</td></tr></table> What would be the optimal scheduling to minimize the total processing time for these two parts? Find also the total elapsed time. For each machine specify the part that should be processed first.	Part P_1	Machine sequence:	C	A	E	F	D	B	Time (hrs):	2	3	4	5	6	1	Part P_2	Machine sequence:	B	A	E	F	C	D	Time (hrs):	3	2	5	3	2	3	10	3	3					
Part P_1	Machine sequence:		C	A	E	F	D	B																															
	Time (hrs):	2	3	4	5	6	1																																
Part P_2	Machine sequence:	B	A	E	F	C	D																																
	Time (hrs):	3	2	5	3	2	3																																

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Quiz	Max Marks	1	2	2	—	1	2	2	—	—	—
	Test	Max Marks	10	20	20	—	10	20	20	—	—	—
